

AI-Powered Resume Screening System

Project Synopsis

Project Title

AI-Powered Resume Screening System

1. Introduction

The AI-Powered Resume Screening System is an intelligent web-based application developed using Python and Streamlit that automates the process of screening and shortlisting resumes based on a given job description.

In traditional recruitment systems, HR teams manually review hundreds of resumes, which is time-consuming, inefficient, and prone to human error. This project solves that problem by automatically extracting resume content, identifying important skills, comparing resumes with job requirements, and ranking candidates according to their matching score.

The system supports multiple resume uploads in PDF and DOCX formats and provides a modern interactive dashboard for recruiters.

2. Objective of the Project

The main objectives of this project are:

- Automate resume screening process
- Reduce manual HR effort
- Improve hiring efficiency
- Identify best matching candidates automatically
- Extract candidate details from resumes
- Analyze technical skills and missing skills
- Generate resume ranking and shortlisting reports

3. Features of the System

Core Features

- Multiple Resume Upload
- PDF Resume Support
- DOCX Resume Support
- Job Description Input
- Resume Text Extraction
- Keyword Matching
- Match Percentage Calculation
- Skill Extraction
- Missing Skill Detection
- Resume Ranking
- Candidate Shortlisting

Advanced Features

- Candidate Name Extraction
- Email Extraction
- Phone Number Extraction
- Score Categorization
- Smart Selection Status
- Dashboard Analytics
- Resume Preview
- CSV Report Download
- Interactive Streamlit UI
- Candidate Filtering System

4. Technologies Used

Technology	Purpose
Python	Backend Logic
Streamlit	Web Application Framework
Pandas	Data Processing
PyPDF2	PDF Text Extraction
python-docx	DOCX File Reading
Regex	Email & Phone Extraction

5. System Architecture



6. Modules of the Project

6.1 Resume Parser Module

This module extracts text content from uploaded PDF and DOCX resumes.

Functions: • PDF Reading • DOCX Reading • Resume Text Extraction

6.2 Skill Extraction Module

This module identifies technical skills present in resumes and job descriptions.

Examples: • Python • SQL • Machine Learning • Streamlit • Django

6.3 Matching Engine Module

This module compares resume content with the job description and calculates matching percentage.

Outputs: • Match Score • Matched Keywords • Missing Skills

6.4 Candidate Information Module

This module extracts:

• Candidate Name • Email Address • Phone Number

using regular expressions and text processing.

6.5 Dashboard Module

Displays:

• Total Resumes • Shortlisted Candidates • Rejected Candidates • Best Matching Resume • Candidate Table

7. Working of the System

Step 1: User uploads multiple resumes.

Step 2: User enters job description.

Step 3: System extracts resume text.

Step 4: Skills are identified from resumes.

Step 5: Resumes are compared with job description.

Step 6: Match score is calculated.

Step 7: Candidates are categorized.

Step 8: Dashboard displays results.

Step 9: Recruiter downloads CSV report.

8. Advantages of the System

• Saves recruitment time • Reduces manual screening effort • Improves hiring accuracy • Handles multiple resumes together • Easy to use interface • Fast candidate filtering • Better recruitment analytics

9. Limitations

• Basic NLP implementation • Skill matching depends on predefined keywords • Resume formatting may affect extraction quality • No database integration in current version

10. Future Enhancements

• AI-based semantic matching • OpenAI/Gemini integration • Interview recommendation system • Resume scoring using Machine Learning • ATS compatibility checking • Database integration • Email automation for recruiters • Candidate analytics dashboard • Authentication system • Cloud deployment

11. Folder Structure

```
AI_Resume_Screening_System/  
|  
├─ app.py  
├─ resume_parser.py  
├─ matcher.py  
├─ skills_extractor.py  
├─ candidate_info.py  
├─ score_category.py  
└─ requirements.txt
```

12. Installation Steps

Step 1: Install Python

Download and install Python from:

<https://www.python.org/downloads/>

Verify installation:

```
python --version
```

Step 2: Create Project Folder

```
mkdir AI_Resume_Screening_System  
cd AI_Resume_Screening_System
```

Step 3: Create Virtual Environment

```
python -m venv venv
```

Activate virtual environment:

Windows

```
venv\Scripts\activate
```

Mac/Linux

```
source venv/bin/activate
```

Step 4: Install Required Libraries

Create requirements.txt

```
streamlit  
pandas  
PyPDF2  
python-docx
```

Install libraries:

```
pip install -r requirements.txt
```

13. How to Run the Project

Step 1

Open terminal inside project folder.

Step 2

Run Streamlit application:

```
streamlit run app.py
```

Step 3

Browser will open automatically.

Default URL:

```
http://localhost:8501
```

14. Input and Output

Input

- Multiple Resume Files
- Job Description

Output

• Match Percentage • Candidate Ranking • Shortlisted Candidates • Missing Skills • Dashboard Analytics • CSV Report

15. Sample Job Description

We are looking for a Python Developer with knowledge of SQL, Machine Learning, Streamlit, Django, Pandas, and API development.

16. Sample Output

Candidate	Match Score	Status
Rahul Sharma	88%	Shortlisted
Aman Kumar	72%	Review Required
Riya Singh	35%	Rejected

17. Conclusion

The AI-Powered Resume Screening System is an efficient recruitment automation solution that simplifies resume analysis and candidate shortlisting. The project demonstrates the practical use of Python, Streamlit, text processing, and skill extraction techniques in real-world HR technology systems.

This system can significantly reduce hiring effort and improve recruiter productivity while providing a scalable foundation for future AI-powered recruitment solutions.

18. Developed By

Name: Dhiraj Kr Profession: GenAI Developer & Data Scientist

👉 End of Document 👈